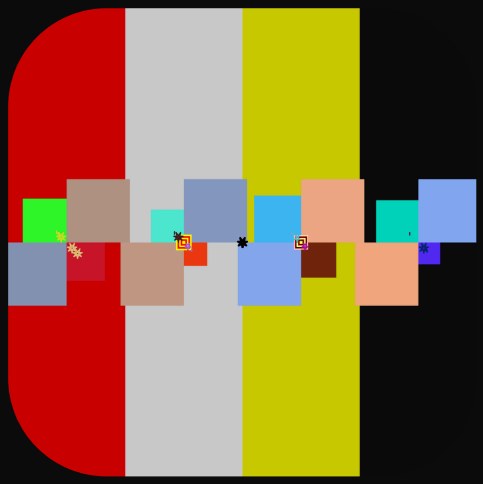


SHOAL

FOUR SYNTHS PLAYING EACH OTHER

USER GUIDE · VERSION 1.0

K.ODK

Four synths in one shoal

Shoal is four synthesisers that play each other. Each one runs a patch written in shlisp — the language of the Shbobo Shnth — or a JUSTINTS just-intonation piece. Wires join them: one hears another and answers it, and the whole shoal moves together.

You do not pick patches from a list. **GEN** grows a new one for any of the four, out of the hundreds of patches inside the app: changed, crossed, grafted, or written word by word from the language itself. Every new patch is heard first and kept only if it really sounds.

You can leave the shoal alone and let it drift, or play it with your fingers on the sixteen keys, on the big picture, and with the phone's tilt or its camera as the Shnth's two antennae.

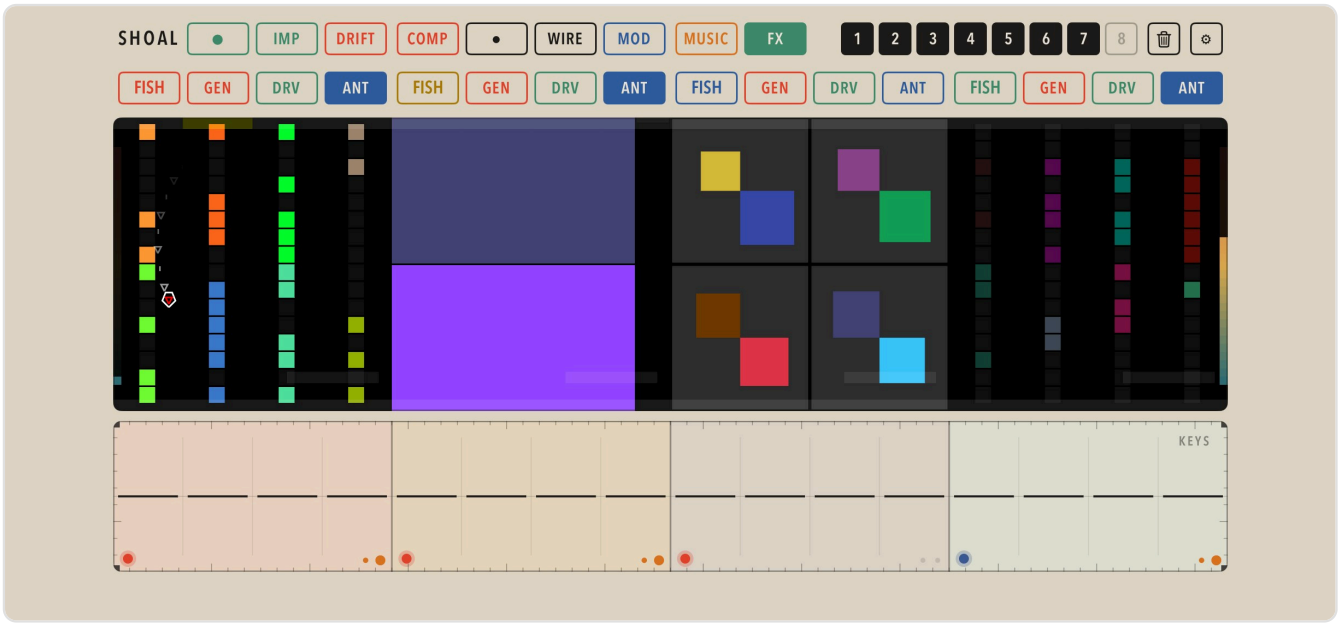


Too much to read? Skip it. Touch the keys along the bottom. Press **GEN** over any of the four for a new sound, **WIRE** to let them hear each other, **IMP** to let them play on their own. Hold a number **1–8** to keep what you hear.

The screen

TOP ROW	The whole shoal: tempo, IMP, DRIFT, COMP, balls, WIRE, MOD, MUSIC, FX, the eight presets, the bin and the settings (⚙).
SECOND ROW	The four, each over its own quarter: what it is (FISH · JUST · TOO), GEN, DRV, ANT.
THE PICTURE	The shoal drawn as one big picture. A finger plays it; two taps change it. MAJOR runs along its top edge, minor along its bottom.
THE KEYS	Sixteen keys, four for each synth, in its colour. What the wires bring each one is written on them.

A, B, C, D



Each of the four has its own quarter of the screen, in its own colour, with four keys on the second row:

FISH	**What it is.** Tap to change it in turn — **FISH** (a Shnth running a shlisp patch) → **JUST** (a JUSTINTS piece) → **TOO** (a justintstoo piece) → FISH. Each change brings a new patch of that kind. **Hold** it to open its panel (see below).
GEN	**A new patch** for this one, grown from the others (red for a Shnth, orange for a JUSTINTS). It lights a moment when the new one comes in. How far it strays is set in the settings.
DRV	**Drive:** its code changes on its own, a little at a time — every few seconds a small change of one kind (a number, a number set moving, its tuning, a source, its motion), each heard before it is kept.
ANT	**The antennae** reach this one. The phone's tilt or the camera (chosen in the settings) plays the Shnth's two antennae, A and B. On A only when you first open the app.

ITS PANEL (HOLD FISH / JUST / TOO)

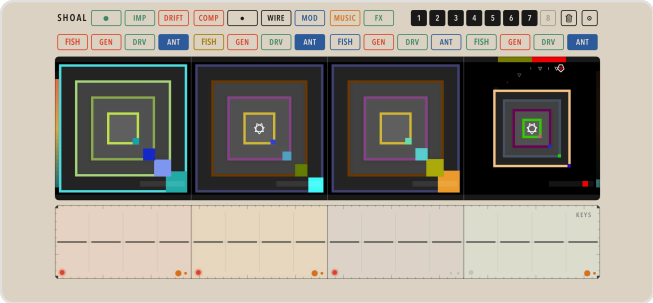
SHNTH | JUSTINTS switches what it is. A JUSTINTS also shows its pad, drawn as justints draws it: **FM** (a star), **ROUTE** (king and queen), **HOLD** (the prime limit held), **RAMP** (ramp or square) and **SAW** (saw or triangle) — tap one to turn it — and the **FM × STEP** square: drag the crossing lines to set FM shift (across) and STEP (up). Tap outside the panel to close it.

The top row

●	**Tempo.** Tap it two or more times to set a tempo; it then blinks on the beat and the IMPs play on its sixteenths. Set it to FREE in the settings to let them play freely.
IMP	**The IMPs** play the keys on their own: none → one → two → three → none. How each plays is set in the settings.
DRIFT	**GEN on its own.** Every so often one of the four is grown anew — sooner when the tide is high.
COMP	**Compose.** While it is on, GEN (and DRIFT) no longer changes existing patches: it writes a new one word by word from the language as the patches use it — every word, its places, its numbers.
•	**The balls:** off → one → two → three. A ball moves through the keys; where it hits the frame above them (MAJOR) or below them (minor), that key's button is pressed for a moment. How it moves is set in the settings.
WIRE	**Who hears whom.** Each press gives every pair one more wire, up to three, then none. New wires each time, tried first and kept where all four answer. Its colour shows how many: green one, orange two, red three.
MOD	**Joined through their code.** Press once: every so often one takes another's motion (a branch of its code) into its own. Press again (green): that, and **DRIVE** — what the shoal is doing written into the code: how loud, bright or slow the others are, the tide, where the ball is. Press again: off.
MUSIC	**All four in one key** — one just scale on one 1/1, chosen at random. Every pitch is set in the key and nothing bends it. Press once: MUSIC (played freely in the key). Again (blue): MUSIC 2, the same key with the pitches walking the scale. Again: off, each back as it came. Whatever comes in while it is on — GEN, DRIFT, a preset — is set in the same key.
FX	**The picture's effects** on the whole sound, on or off. Each picture has its own chain (below).
1 … 8	**Presets.** Hold a number to keep the whole shoal there (it flashes). Tap it to bring it back. The one playing is orange, kept ones black, empty ones faint.
BIN	Light the bin, then tap a preset to clear it.
⚙	The settings, over the whole screen, each with a line on what it does.

The tide is a slow movement of its own, wandering under everything. It drifts the ball, makes DRIFT and MOD come sooner, and can be wired like one of the four.

The picture and the keys





The picture

Two quick taps change the picture: ALL · RUNE · FIRST · SQUARE · STAR · FLAG · NEST · BIT · PRIME · STEP (and TOO, justintstoo's own). **A finger on it** plays the picture's effect — across its first setting, up its second — and plays the shoal itself, each picture its own way:

ALL · NEST	The pitch of the one under the finger.
RUNE	A hand at that one's antennae.
FIRST	The balls are drawn to the finger.
SQUARE · BIT	Its buttons: the upper half MAJOR, the lower half minor, under the finger.
STAR · PRIME	That one shaken by its neighbour (FM).
FLAG · STEP	The key under the finger, as high as the finger is.

The thin frames along the picture's top (**MAJOR**) and bottom (**minor**) are buttons, one under each key. Touch them to hold a button; the balls press them as they hit. The antennae show as small squares at the picture's left (A) and right (B) edges.

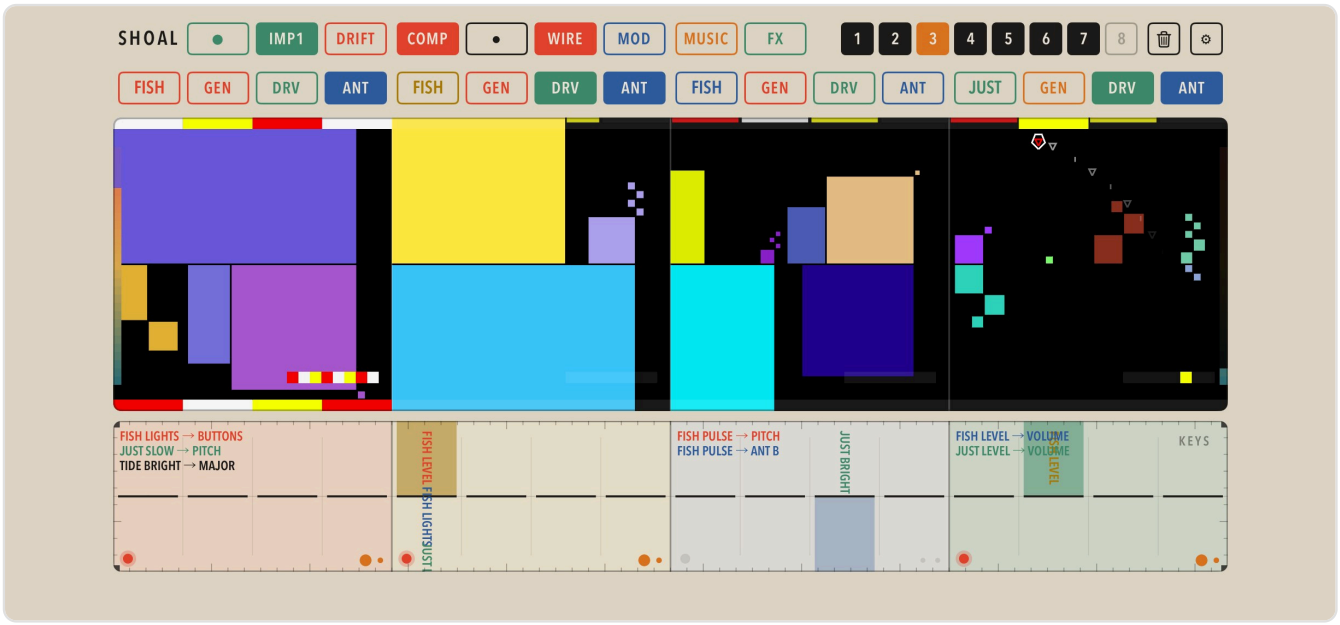
The keys

Sixteen keys, four for each synth — its four bars. Play many at once. A key comes up the moment you touch it: the higher your finger, the more, and a flat finger more than a fingertip. The upper half bends it up, the lower half down. Slide sideways onto the next keys. Let go and it falls back along a slope, as a piezo does.

Double tap a key to hold it where it is (a mark appears on top; touched again, it moves and stays held). Double tap again to let it go. On a JUST's keys, a double tap in the upper half copies its voice and in the lower half takes one away.

The filled bars show what each key really gets — your fingers, the IMPs and the wires together. The dot at the lower left of each quarter is its output level (blue quiet, red loud); the two at the lower right are its antennae.

Wires



A wire takes something from one of the four — or from the tide — to another. What the wires bring each one is written on its keys: over a bar, who moves that bar; along the top, the rest (for example *FISH PULSE → PITCH*).

WHAT IS HEARD	
LEVEL	how loud it is
BRIGHT	how bright
PULSE	when it starts a sound
SLOW	how loud of late
LIGHTS	its eight LEDs
STEP	which preset of its patch it is on
SOUND	its sound itself (for FM)

WHERE IT GOES	
BAR 1–4 · ANT A/B	its hands, as yours would
MINOR · MAJOR	a button
BUTTONS	all eight at once
TAR · NEXT	TAR pressed · its next preset
VOLUME · PAN	how loud · where it stands
RING	its sound times the other's
FM · PITCH	its pitch shaken at audio rate · bent

The effects

With **FX** on, each picture brings its own effect chain over the whole sound, one of them strong — ALL all fourteen, RUNE lo-fi and glitch, FIRST grains and room, SQUARE fold, STAR frequency shift and ring, FLAG tape and filter, NEST room, BIT glitch, PRIME harmoniser, STEP pitch. A finger on the picture plays it.



GEN	How far GEN strays: 1 stays near the patches' own, 12 is chaos. AUTO picks anew each time.
ANT	Where the antennae come from: **TILT** (tipping the phone away / towards is A, turning it like a wheel is B) · **CAM** (the camera as two light sensors — a hand over its upper half darkens A, over its lower half B) · **BALL** (where the first ball is) · **OFF**. Tap CAM again for the other camera. Hold any of them to take the present position or light as rest (zero).
CAM	FRONT or BACK camera.
TEMPO	The IMPs' tempo, or FREE.
BALL	How the ball moves: **DRF** kicked by sounds and drawn by keys, drifting · **BNC** a billiard ball that never slows · **CHS** after the loudest one · **ORB** round and round, as fast as they sound · **PUL** drawn to each as loud as it is · **HND** after your fingers (let go, it falls).
WANDER · PULL	For DRF: how much the ball roams on its own, how much a pressed key draws it.
IMP 1–3	Each IMP its way — **LONG** long and seldom, **BUSY** short and many, **SLIDE** gliding to the next key, **CHORD** a few at once, **MIX** any. **OFTEN**: how often it plays. **HOLD**: how long it holds a key.

Notes

The shoal you leave is still there the next time you open the app. Shoal keeps playing in the background. The camera is opened only if you choose CAM, and is read only as light; no image is kept. Nothing is collected.

Credits

The Shnth's sound engine (a C port of its firmware), the shlisp compiler, justints, justintstoo and the shbobo patches in Shoal: Peter Blasser, github.com/pblasser/shbobo — MIT License. Further patches: montoyamoraga, shbobo-studies — MIT License. The full licence text is in the app's settings. Shoal is not made or endorsed by Shbobo or Ciat-Lonbarde.

Support: keiodak.github.io/privacy/shoal-support.html